

WEEK-1

OUT-PUT EXERCISE-7

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:E:\IntelliJ2026\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=6258
Present Value = 10000.0
Growth Rate = 0.1
Years = 5

Future Value = 16105.100000000008

Process finished with exit code 0
```

Explanation:

Time Complexity: $O(n)$

Space Complexity: $O(n)$

Optimization Techniques

1. Memoization

- Store previously computed results and reuse them.

2. Iterative Approach

- An iterative solution eliminates recursive overhead.

2. Iterative Approach

- An iterative solution eliminates recursive overhead.